

1. Title

ContractIQ: Contract Obligation Tracking & Compliance Management Platform

2. Objective

Build a full-stack web application using Angular and FastAPI that enables organizations to securely manage contracts, track contractual obligations, monitor renewal dates, analyze compliance status, and receive proactive notifications for important legal milestones.

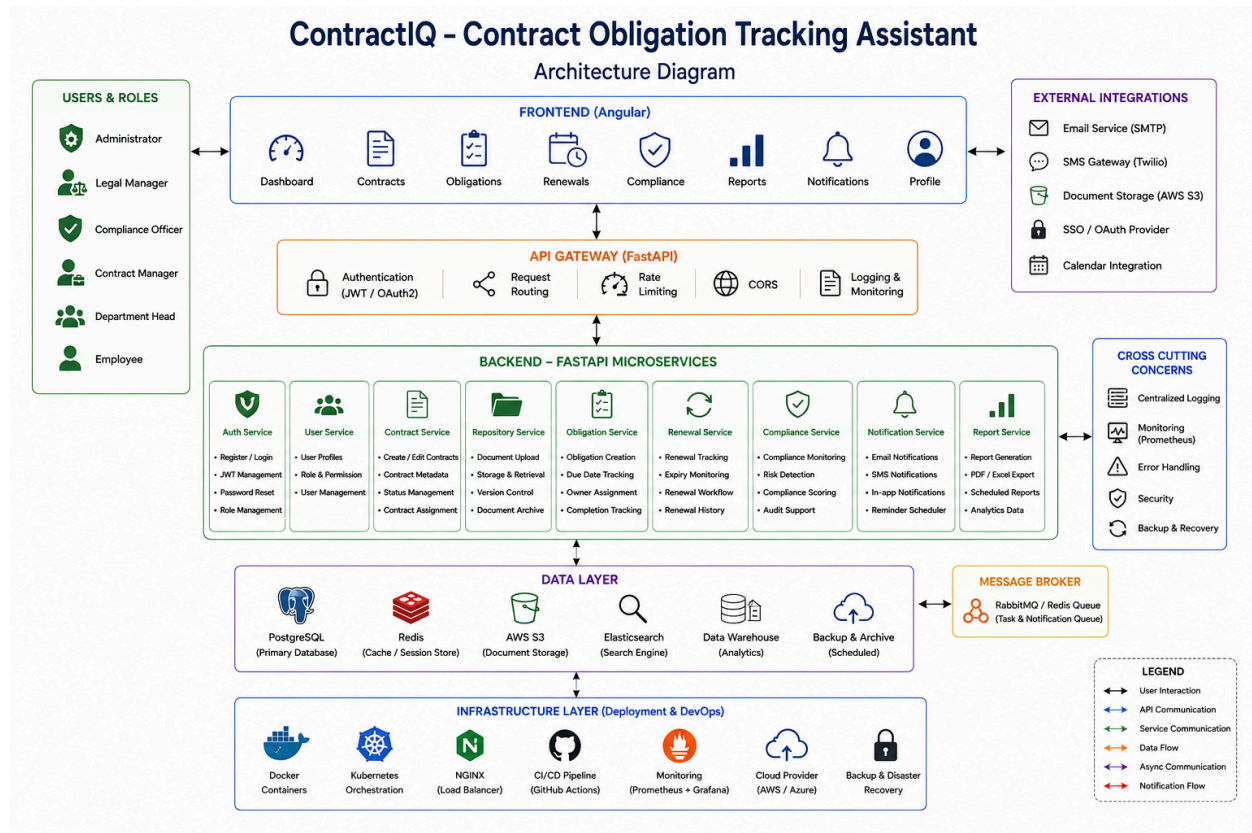
The platform helps legal teams, procurement departments, HR teams, and business organizations reduce compliance risks by centralizing contract management, obligation tracking, document storage, renewal monitoring, and reporting.

This solution can be used by enterprises, law firms, procurement departments, healthcare organizations, educational institutions, government agencies, and startups.

Outcomes

- (i) Developed and deployed a full-stack Contract Management platform.
- (ii) Implemented secure user authentication and role-based access control.
- (iii) Built contract repository and document management modules.
- (iv) Developed obligation tracking and compliance monitoring workflows.
- (v) Implemented renewal reminder and notification systems.
- (vi) Built contract approval and review workflows.
- (vii) Developed dashboards for contract analytics and compliance monitoring.
- (viii) Implemented document version control and audit logging.
- (ix) Generated contract compliance and renewal reports.
- (x) Deployed the application using Docker and cloud platforms.
- (xi) Designed responsive UI wireframes and legal workflow screens before implementation.

3. Architecture Components



4. Modules to be Implemented

1. User Authentication & Role Management

Features

- (i) User Registration
- (ii) Secure Login
- (iii) JWT Authentication
- (iv) Password Reset
- (v) Profile Management
- (vi) Role-Based Access Control

Roles

Administrator

Legal Manager

Compliance Officer

Contract Manager

Department Head

Employee

2. Contract Repository Module

Features

(i) Contract Upload

(ii) Contract Storage

(iii) Document Categorization

(iv) Document Search

(v) Version Management

(vi) Document Archive

Contract Categories

Employment Contracts

Vendor Contracts

Service Agreements

Lease Agreements

Purchase Agreements

Partnership Agreements

Confidentiality Agreements

3. Contract Management Module

Features

- (i) Contract Creation
- (ii) Contract Editing
- (iii) Contract Approval Workflow
- (iv) Contract Review
- (v) Contract Assignment
- (vi) Contract Status Management

Contract Status

Draft

Under Review

Approved

Active

Expired

Terminated

4. Obligation Tracking Module

Features

- (i) Obligation Registration
- (ii) Due Date Tracking
- (iii) Responsibility Assignment
- (iv) Completion Status
- (v) Compliance Monitoring
- (vi) Progress Tracking

Obligation Types

Payment Obligations

Delivery Commitments

Reporting Requirements

Renewal Conditions

Service Level Agreements

Legal Compliance Requirements

5. Renewal Management Module

Features

(i) Renewal Tracking

(ii) Expiry Monitoring

(iii) Auto Reminder Scheduling

(iv) Renewal Approval Workflow

(v) Renewal History

(vi) Renewal Status

Renewal Status

Upcoming

In Progress

Renewed

Expired

Cancelled

6. Compliance Monitoring Module

Features

- (i) Compliance Status Tracking
- (ii) Missed Obligation Detection
- (iii) Compliance Reports
- (iv) Risk Identification
- (v) Audit Support
- (vi) Compliance History

Compliance Levels

Compliant

Pending

Delayed

Non-Compliant

High Risk

7. Dashboard & Analytics Module

Legal Dashboard

- (i) Active Contracts
- (ii) Upcoming Renewals
- (iii) Pending Obligations
- (iv) Compliance Status
- (v) Recent Activities

Compliance Dashboard

- (i) Compliance Reports
- (ii) Missed Deadlines
- (iii) Risk Indicators

- (iv) Audit Summary
- (v) Department Performance

Admin Dashboard

- (i) User Management
- (ii) Contract Statistics
- (iii) System Monitoring
- (iv) Notification Status
- (v) Activity Logs

8. Notification Module

Features

- (i) Renewal Reminders
- (ii) Obligation Due Alerts
- (iii) Compliance Notifications
- (iv) Contract Approval Alerts
- (v) Email Notifications
- (vi) SMS Notifications
- (vii) In-App Notifications

9. Reports & Export Module

Features

- (i) Contract Reports
- (ii) Compliance Reports
- (iii) Renewal Reports
- (iv) Obligation Reports

(v) Audit Reports

(vi) PDF Export

(vii) Excel Export

10. Audit & Activity Management

Features

(i) User Activity Logs

(ii) Contract History

(iii) Approval Logs

(iv) Audit Reports

(v) Security Logs

(vi) Change History

11. Final Integration, Testing & Deployment

5. Milestone 1: Week 1 & 2 — Requirements, UI Design, Database Design & Backend Setup

Tasks

(i) Define project objectives.

(ii) Gather functional and non-functional requirements.

(iii) Design UI wireframes.

(iv) Create user flow diagrams.

(v) Prepare dashboard layouts.

(vi) Design responsive screens using Figma.

(vii) Design database schema.

Tables

users

contracts

contract_versions

obligations

renewals

notifications

reports

audit_logs

activities

(viii) Initialize FastAPI project.

(ix) Configure PostgreSQL.

(x) Implement JWT Authentication.

(xi) Create Angular frontend.

(xii) Configure Angular Material.

UI Wireframes

Login Page

Registration Page

Dashboard

Contract Repository

Contract Details

Obligation Tracker

Renewal Dashboard

Compliance Dashboard

Reports Dashboard

Notification Screen

Outcomes

- (i) UI wireframes completed.
- (ii) User flows finalized.
- (iii) Database schema completed.
- (iv) FastAPI backend initialized.
- (v) Angular frontend initialized.
- (vi) Authentication implemented.

6. Milestone 2: Week 3 & 4 — Contract & Repository Management

Tasks

- (i) Develop Contract Repository.
- (ii) Build Contract Management module.
- (iii) Implement Version Control.
- (iv) Create Approval Workflow.
- (v) Develop Search Functionality.
- (vi) Build Document Archive.

Outcomes

- (i) Contract Repository operational.
- (ii) Contract workflows functional.
- (iii) Version Management completed.

(iv) Approval workflow operational.

7. Milestone 3: Week 5 & 6 — Obligation Tracking & Compliance Monitoring

Tasks

- (i) Develop Obligation Tracking module.
- (ii) Build Renewal Management system.
- (iii) Implement Compliance Monitoring.
- (iv) Create Notification System.
- (v) Generate Reports.
- (vi) Build Analytics Dashboard.

Outcomes

- (i) Obligation tracking operational.
- (ii) Compliance monitoring completed.
- (iii) Notification system functional.
- (iv) Reports generated successfully.

8. Milestone 4: Week 7 & 8 — Testing, Deployment & Documentation

Tasks

- (i) Perform application testing.
- (ii) Fix identified issues.
- (iii) Deploy frontend and backend.
- (iv) Configure Docker.

- (v) Prepare documentation.
- (vi) Demonstrate complete application.

Outcomes

- (i) Production-ready application.
- (ii) Deployment completed.
- (iii) Documentation completed.
- (iv) Successful project demonstration.

9. Evaluation Criteria

Milestone 1

- (i) FastAPI project setup completed.
- (ii) Angular application initialized.
- (iii) Database schema finalized.
- (iv) Authentication implemented.
- (v) UI wireframes completed.

Milestone 2

- (i) Contract Repository operational.
- (ii) Contract workflows functional.
- (iii) Version Control completed.
- (iv) Search functionality operational.

Milestone 3

- (i) Obligation Tracking operational.
- (ii) Compliance Monitoring functional.

(iii) Analytics Dashboard completed.

(iv) Reports generated successfully.

Milestone 4

(i) Application deployed successfully.

(ii) Testing completed.

(iii) Documentation prepared.

(iv) Complete workflow demonstrated.

10. Tools & Tech Stack

Programming Language

Backend

Python

FastAPI

Frontend

TypeScript

Angular

Database

PostgreSQL

Redis

Backend Framework

FastAPI

SQLAlchemy

Pydantic

Alembic

Uvicorn

Celery

Frontend Framework

Angular

Angular Material

Bootstrap

RxJS

Chart.js

Authentication

JWT Authentication

OAuth2

Role-Based Access Control

Notifications

SMTP Email

Twilio SMS

Firebase Cloud Messaging

Cloud & DevOps

Docker

Docker Compose

GitHub Actions

AWS

Azure

Nginx

Development Tools

VS Code

Git

GitHub

Postman

Figma

11. Performance Metrics

Contract Management

Contract Upload Success Rate

Contract Approval Time

Document Retrieval Time

Compliance Performance

Obligation Completion Rate

Renewal Tracking Accuracy

Compliance Status Accuracy

Notification Performance

Reminder Delivery Success Rate

Alert Response Time

Notification Processing Time

System Performance

API Response Time

Database Query Performance

Dashboard Loading Time

Concurrent User Handling

12. Example Quantitative Goals

Contract Upload Time less than 5 seconds

Contract Search Results below 2 seconds

95% Obligation Tracking Accuracy

API Response Time below 300 ms

Dashboard Load Time below 2 seconds

Support 1000+ Concurrent Users